

Assignment 1

(Sep. 14-21)

Problem 1:

Waterloo's infamous Maple Syrup Madness produces maple syrup for the entire southern part of Ontario. The following table shows the demand d_i (in tons) for each month $i \in \{1, \dots, 12\}$.

Month	1	2	3	4	5	6	7	8	9	10	11	12
d_i	10	12	14	20	15	18	19	20	30	28	31	22

The company needs to decide how much maple syrup to produce in each month so that the demand can be met. As usual we make the following assumption: maple syrup produced in month i can be consumed in month i (and later, of course).

- (a) Assume that producing a ton of maple syrup in month i costs p_i , and that the company can store up to 15 tons of its product in its storage facility for an unlimited amount of time. Assume that the storage facility contain 7 tons of maple syrup at the beginning of month 1.

The company wants to determine how much to produce in each month in order to minimize the total production cost. Write an LP for this task.

Solution: Let x_i be the amount of maple syrup produced in month i , and let s_i be the number of tons stored at the end of the month.

$$\begin{aligned}
 \min \quad & \sum_{i=1}^{12} p_i x_i \\
 \text{s.t.} \quad & x_i + s_{i-1} - d_i = s_i \quad (1 \leq i \leq 12) \\
 & s_i \leq 15 \quad (1 \leq i \leq 12) \\
 & s_0 = 7 \\
 & x, s \geq 0
 \end{aligned}$$

- (b) Now assume that the company wants to even out production over the year in order to keep its demand for staff constant. We model this by assuming that the company incurs a cost of 50 for each ton of difference between the production quantities in any two consecutive months. As an example, assume that the company decides to produce x_i tons of maple syrup in month $i \in \{1, \dots, 12\}$.

i	1	2	3	4	5	6	7	8	9	10	11	12
x_i	4	19	21	16	32	32	20	10	12	5	6	8

The cost incurred by the company for the production plan given in the above table is:

$$\begin{aligned}
 & ((19 - 4) + (21 - 19) + (21 - 16) + (32 - 16) + (32 - 32) + (32 - 20) + \\
 & (20 - 10) + (12 - 10) + (12 - 5) + (6 - 5) + (8 - 6)) \cdot 50.
 \end{aligned}$$

As in (a), assume that the storage facility contains 7 tons of maple syrup at the beginning of month 1, and that you can store up to 15 tons of maple syrup at any time.

The company wants to find a production schedule that minimizes the total cost incurred by production changes. Formulate the problem as an LP.

Solution: As before, we let x_i be the production quantity in month $1 \leq i \leq 12$, and let s_i be the storage requirement at the end of month i . We want to minimize

$$\sum_{i=1}^{11} |x_i - x_{i+1}| 50.$$

We need to linearize the absolute value in the above expression, and do this by introducing new variables z_i with intended meaning $|x_i - x_{i+1}|$ for all i . We enforce this meaning by adding two constraints for each i :

$$\begin{aligned} z_i &\geq x_i - x_{i+1} \\ z_i &\geq x_{i+1} - x_i. \end{aligned}$$

Clearly if z_i is equal to $|x_i - x_{i+1}|$ then it satisfies these two constraints. On the other hand, if z_i is minimal and satisfies these two constraints then $z_i = |x_i - x_{i+1}|$. With this the new LP becomes:

$$\begin{aligned} \min \quad & 50 \sum_{i=1}^{11} z_i \\ \text{s.t.} \quad & x_i + s_{i-1} - d_i = s_i \quad (1 \leq i \leq 12) \\ & s_i \leq 15 \quad (1 \leq i \leq 12) \\ & s_0 = 7 \\ & z_i \geq x_i - x_{i+1} \quad (1 \leq i \leq 11) \\ & z_i \geq x_{i+1} - x_i \quad (1 \leq i \leq 11) \\ & x, s, z \geq 0 \end{aligned}$$

Problem 2:

Consider the following word game. You are assigned a number of tiles, each containing letters from the alphabet $\Sigma = \{a, b, c, \dots, z\}$. Suppose that your input contains $N_\alpha \in \mathbb{N}$ tiles with the letter α for each $\alpha \in \Sigma$. From the letters you are to construct any of the words $w_1, \dots, w_n$. This list might for example consist of all the words in a dictionary.

You must construct any word at most once, and use any tile at most once. You receive $v_j \geq 0$ points for constructing word w_j , and an additional bonus $b_{ij} \geq 0$ for making both words w_i and w_j .

- (a) Formulate an integer program to determine your optimal choice of words (one that maximizes the total number of points + bonus).

Solution: For simplicity, let us define $w_{i,\alpha}$ for all $i \in [n]$ and $\alpha \in \Sigma$ as the number of times letter α occurs in word w_i . Notice that w_i is a constant that depends only on the input to the question! It is not a variable.

We now define a binary variable x_i for all $i \in [n]$ that we want to have value 1 if word w_i is chosen, and 0 otherwise. Define a binary variable variable x_{ij} that will take value 1 if both w_i and w_j are chosen. The MIP is as follows:

$$\begin{aligned} \max \quad & \sum_{j \in [n]} v_j x_j + \sum_{i,j \in [n]} b_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i \in [n]} w_{i,\alpha} x_i \leq N_\alpha \quad \forall \alpha \in \Sigma \end{aligned} \quad (1)$$

$$\begin{aligned} x_{ij} &\leq x_i \quad \forall i, j \in [n], \forall l \in \{i, j\} \\ x_i, x_{ij} &\in \{0, 1\} \forall i, j \in [n] \end{aligned} \quad (2)$$

Constraint (1) says that the total number of α characters in the words chosen is no larger than then the tiles N_α that are available for this letter. Constraint (2) states that x_{ij} must be 0 if at least one of words w_i and w_j is not chosen. When both are chosen then x_{ij} will be set to 1 in an optimal solution as long as b_{ij} is positive. The objective function measures correctly the value of formed words plus bonus.

- (b) How does your formulation change when you are allowed to select 100 tiles with no restriction on your choice of letters?

Solution: Essentially, we now make N_α a variable for all $\alpha \in \Sigma$. To avoid confusion, let n_α be the number of tiles with letter $\alpha \in \Sigma$ that we decide to choose. We then have the following MIP:

$$\begin{aligned} \max \quad & \sum_{j \in [n]} v_j x_j + \sum_{i,j \in [n]} b_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i \in [n]} w_{i,\alpha} x_i \leq n_\alpha \quad \forall \alpha \in \Sigma \end{aligned} \quad (3)$$

$$x_{ij} \leq x_i \quad \forall i, j \in [n], \forall l \in \{i, j\} \quad (4)$$

$$\sum_{\alpha \in \Sigma} n_\alpha \leq 100 \quad (5)$$

$$\begin{aligned} x_i, x_{ij} &\in \{0, 1\}, \forall i, j \in [n] \\ N_\alpha &\geq 0 \text{ and integer}, \forall \alpha \in \Sigma \end{aligned}$$

Constraint (5) forces feasible solutions to choose at most 100 tiles. All other constraints are as before.

Problem 3:

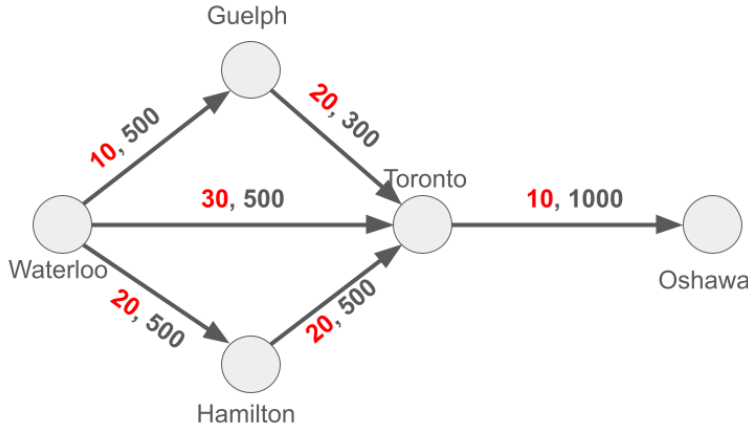


Figure 1: Maple Syrup Madness Pipe Network for Problem 3. The numbers **10**, 500 on the arc connecting Waterloo to Guelph signal that it costs 10 cents to send a litre from Waterloo to Guelph, and that it is possible to send up to 500 litres.

Maple Syrup Madness has a plant in each of the cities of Waterloo, Guelph, Hamilton, Toronto, and Oshawa. Maple Syrup Madness marketing team has proposed that selling maple syrup for 100 cents in Oshawa may be a good idea for the company brand. To attend this demand, the company will employ its remarkable pipe infrastructure connecting its plants — see Figure 1. Namely:

- the Waterloo plant can send maple syrup to Guelph, Hamilton, and Toronto, at the cost of 10, 20, and 30 cents per litre, respectively;
- the Guelph plant can send maple syrup to Toronto at the cost of 20 cents per litre;
- the Hamilton plant can send maple syrup to Toronto at the cost of 20 cents per litre;
- the Toronto plant can send maple syrup to Oshawa at the cost of 10 cents per litre.

The Waterloo plant has a total of 1000 litres of maple syrup that can be used in this publicity stunt. But given the short window between when these litres are produced and when they must be sold in Oshawa, the pipe capacities impose restrictions on how many litres can be sent in each pipe. Each of the pipes listed above has a maximum capacity of 500 litres; except the pipe between Guelph and Toronto, that can only be used to send 300 litres, and the recently upgraded pipe between Toronto and Oshawa, which can be used to send 1000 litres.

The company would like to know the maximum profit it can achieve from selling Waterloo's maple syrup in Oshawa for 100 cents. Formulate an LP to solve that task.

Solution:

We have 6 variables: $x_{\vec{wg}}$, $x_{\vec{wh}}$, $x_{\vec{wt}}$, $x_{\vec{gt}}$, $x_{\vec{ht}}$, and $x_{\vec{to}}$. The variable $x_{\vec{wg}}$ encodes how many litres will be sent from Waterloo to Guelph, and similarly for the other variables. Our objective

function is

$$100x_{\vec{to}} - (10x_{\vec{wg}} + 20x_{\vec{wh}} + 30x_{\vec{wt}} + 20x_{\vec{gt}} + 20x_{\vec{ht}} + 10x_{\vec{to}}),$$

which subtracts, from the profit of selling the maple syrup in Oshawa, the costs of shipping it there. Each variable must be bounded by the capacity of the relevant arc:

$$\begin{aligned} x_{\vec{wg}} &\leq 500, & x_{\vec{wh}} &\leq 500, & x_{\vec{wt}} &\leq 500, \\ x_{\vec{gt}} &\leq 300, & x_{\vec{ht}} &\leq 500, & x_{\vec{to}} &\leq 1000. \end{aligned}$$

For each plant which is not Waterloo or Oshawa, we need to impose that all the maple syrup that enters equals the maple syrup that leaves:

$$x_{\vec{wg}} - x_{\vec{gt}} = 0, \quad x_{\vec{wh}} - x_{\vec{ht}} = 0, \quad x_{\vec{wt}} + x_{\vec{ht}} + x_{\vec{gt}} - x_{\vec{to}} = 0.$$

We impose that at most 1000 litres leave Waterloo:

$$x_{\vec{wg}} + x_{\vec{wh}} + x_{\vec{wt}} \leq 1000.$$

Finally, we also require all our variables to be nonnegative:

$$x_{\vec{wg}}, x_{\vec{wh}}, x_{\vec{wt}}, x_{\vec{gt}}, x_{\vec{ht}}, x_{\vec{to}} \geq 0.$$

Problem 4: (Review of linear algebra)

(a) Show that the vectors

$$\mathbf{A}_1 := \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \mathbf{A}_2 := \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \mathbf{A}_3 := \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

form a basis for $\mathbb{R}^3$.

(b) Express each of the following vectors as linear combinations of $\mathbf{A}_1$, $\mathbf{A}_2$, and $\mathbf{A}_3$:

$$\mathbf{A}_4 := \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \mathbf{A}_5 := \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \mathbf{A}_6 := \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

(c) Let m and k be a pair of positive integers.

- Let $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k\} \subset \mathbb{R}^m$ be a linearly independent subset of vectors. Suppose $\mathbf{A}_{k+1} \in \mathbb{R}^m$ is not in the subspace spanned by the vectors $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k\}$. Prove that the set $\{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k, \mathbf{A}_{k+1}\}$ is linearly independent.

(d) Prove that every linearly independent subset of vectors in $\mathbb{R}^m$ is part of a basis. So, given a linearly independent subset of vectors (say k of them) in $\mathbb{R}^m$, we can always find $m - k$ vectors to extend the initial subset to a basis for $\mathbb{R}^m$.

Solution: Let

$$\mathbf{A}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{A}_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{A}_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

(a) Form the matrix whose columns are $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$:

$$M = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 2 & -1 \\ 1 & 1 & 1 \end{bmatrix}.$$

Its determinant is

$$\begin{aligned} \det(M) &= 1 \begin{vmatrix} 2 & -1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & -1 \\ 1 & 1 \end{vmatrix} - 1 \begin{vmatrix} 0 & 2 \\ 1 & 1 \end{vmatrix} \\ &= 1(2 + 1) - 1(1) - 1(-2) \\ &= 4. \end{aligned}$$

Since $\det(M) \neq 0$, the columns of M are linearly independent. Thus, $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ are three linearly independent vectors in $\mathbb{R}^3$. Therefore,

$$\boxed{\{\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3\} \text{ is a basis for } \mathbb{R}^3.}$$

(b) To express a vector $\mathbf{v} \in \mathbb{R}^3$ as a linear combination of $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$, we seek scalars c_1, c_2, c_3 such that

$$M \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \mathbf{v}.$$

Solving this system for each of the three standard basis vectors gives the following expressions.

For $\mathbf{A}_4 = (1, 0, 0)^T$,

$$\boxed{\mathbf{A}_4 = \frac{3}{4}\mathbf{A}_1 - \frac{1}{4}\mathbf{A}_2 - \frac{1}{2}\mathbf{A}_3.}$$

For $\mathbf{A}_5 = (0, 1, 0)^T$,

$$\boxed{\mathbf{A}_5 = -\frac{1}{2}\mathbf{A}_1 + \frac{1}{2}\mathbf{A}_2.}$$

For $\mathbf{A}_6 = (0, 0, 1)^T$,

$$\boxed{\mathbf{A}_6 = \frac{1}{4}\mathbf{A}_1 + \frac{1}{4}\mathbf{A}_2 + \frac{1}{2}\mathbf{A}_3.}$$

(c) Suppose that $\{\mathbf{A}_1, \dots, \mathbf{A}_k\} \subseteq \mathbb{R}^m$ is linearly independent and that

$$\mathbf{A}_{k+1} \notin \text{Span}\{\mathbf{A}_1, \dots, \mathbf{A}_k\}.$$

Suppose that

$$c_1\mathbf{A}_1 + \dots + c_k\mathbf{A}_k + c_{k+1}\mathbf{A}_{k+1} = \mathbf{0}.$$

We first show that $c_{k+1} = 0$. If $c_{k+1} \neq 0$, then we could rearrange the equation to obtain

$$\mathbf{A}_{k+1} = -\frac{c_1}{c_{k+1}}\mathbf{A}_1 - \cdots - \frac{c_k}{c_{k+1}}\mathbf{A}_k.$$

This would imply that $\mathbf{A}_{k+1} \in \text{Span}\{\mathbf{A}_1, \dots, \mathbf{A}_k\}$, contradicting the assumption. Hence $c_{k+1} = 0$.

The original equation now reduces to

$$c_1\mathbf{A}_1 + \cdots + c_k\mathbf{A}_k = \mathbf{0}.$$

Because $\{\mathbf{A}_1, \dots, \mathbf{A}_k\}$ is linearly independent, we must have

$$c_1 = \cdots = c_k = 0.$$

Therefore, all coefficients in the original linear combination are zero, and so

$$\boxed{\{\mathbf{A}_1, \dots, \mathbf{A}_k, \mathbf{A}_{k+1}\} \text{ is linearly independent.}}$$

(d) Let

$$S_0 = \{\mathbf{A}_1, \dots, \mathbf{A}_k\} \subseteq \mathbb{R}^m$$

be linearly independent. Since a linearly independent subset of $\mathbb{R}^m$ has at most m vectors, $k \leq m$.

If $\text{Span}(S_0) = \mathbb{R}^m$, then S_0 is already a basis. In this case, necessarily $k = m$, and no additional vectors are needed.

Suppose instead that $\text{Span}(S_0) \neq \mathbb{R}^m$. Let $\mathbf{A}_{k+1}$ be a vector that is not in $\text{Span}(S_0)$. By part (c), the new set $\{\mathbf{A}_1, \dots, \mathbf{A}_{k+1}\}$ is linearly independent.

Continue this process. Each step adds one vector and preserves linear independence. Because no linearly independent subset of $\mathbb{R}^m$ can contain more than m vectors, the process must stop after exactly $m - k$ additions. When it stops, the resulting set spans $\mathbb{R}^m$ and remains linearly independent; hence it is a basis. This completes the proof.